

Matching with Long-Term Employment Relationships

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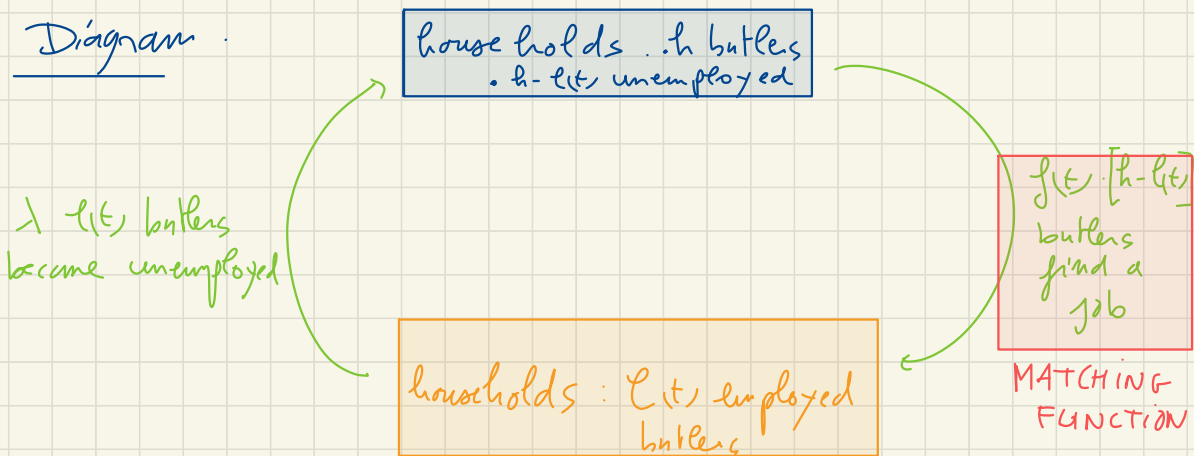
Notation.

- $h > 0$: labor force
- $l(t)$: employment
- $v(t)$: vacancies
- $m(t)$: # new matches at t
- $f(t)$: job-finding rate
- $q(t)$: recruiting rate
- $\lambda > 0$: job-separation rate

Framing of matching market market for butlers

- butlers produce services consumed by households
- each butler produces $a > 0$ services per unit time
- butlers are in long-term employment relationships w/ households

Diagram



Matching function

Cobb-Douglas matching function

2 arguments . - $v(t)$: # vacancies posted by households

- $h - l(t)$ # unemployed workers

matches per unit time,

$$m(t) = \mu v(t)^{1-\eta} [h - l(t)]^\eta$$

$\mu > 0$: matching efficacy

η : matching elasticity (elasticity of matching function wrt unemployment)

$$\eta = d \ln m(t) / d \ln (h - l(t))$$

Matching rates:

- Market tightness:

$$\theta(t) = \frac{v(t)}{h - l(t)}$$

- Job-finding rate, $f(t) = \frac{m(t)}{h - l(t)} = \mu \cdot \theta(t)^{1-\eta}$

- Recruiting rate, $q(t) = \frac{m(t)}{v(t)} = \mu \cdot \theta(t)^{-\eta}$

$\uparrow$
 $q(\theta(t))$